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RESEARCH INTERESTS	Machine Learning, Reinforcement Learning, Multi-Agent Reinforcement Learning	
EDUCATION	Middle East Technical University (METU) , Ankara, Turkey Ph.D. , Computer Engineering, November 2019 • Topic: <i>Improving Reinforcement Learning using Distinctive Clues from the Environment</i> • CGPA: 3.93 / 4.00 • Advisor: Prof. Dr. Faruk Polat , Co-Advisor: Dr. Erkin Çilden M.Sc. , Computer Engineering, August 2016 • GPA: 3.79 / 4.00 • Topic: <i>Effective Subgoal Discovery and Option Generation in Reinforcement Learning</i> • Advisor: Prof. Dr. Faruk Polat B.Sc. , Computer Engineering, June 2014 • GPA: 3.36 / 4.00, HONOR 7 Semesters, HIGH HONOR 1 Semester • Erasmus Program : Wrocław Uni. of Technology, Wrocław, Poland, 2013	
ACADEMIC EXPERIENCE	Assistant Prof. February 2020 - ... Department of Computer Engineering, Izmir University of Economics Postdoctoral Researcher August 2024 - August 2025 Autonomous Agents Research Group, University of Edinburgh Research & Teaching Assistant September 2014 - November 2019 Department of Computer Engineering, Middle East Technical University Supervisor: Prof. Dr. Faruk Polat , Co-Supervisor: Dr. Erkin Çilden	
INDUSTRY EXPERIENCE	Software Developer, BTT Inc., METU Technopolis 2013-2014 - Developing applications on Android - Involving in Web Development Intern, Turkish Aerospace Industries (TAI) Summer 2012 - Working with senior developers in designing and developing software tools - Writing tests for Human Machine Interface partition	
ACADEMIC SERVICE	Conference Reviewing - Association for the Advancement of Artificial Intelligence (2022) - European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (2021, 2022) - International Informatics and Software Engineering Conference (2022) - Multi-disciplinary Conference on Reinforcement Learning and Decision Making (2025) - International Conference on Emerging Trends and Applications in Artificial Intelligence (2026) Journal Reviewing - Applied Intelligence (2023, 2024) - Artificial Intelligence Review (2024) - International Journal of Machine Learning and Cybernetics (2024) - Artificial Intelligence Communications (2024) - Knowledge-based Systems (2024) - Machine Learning (2024)	
PROJECTS	- Academic Advisor, <i>Developing an AI-Backed Personal Management System Based on Real-Time Data for Music Artists</i> , University-Industry Collaboration Project, 2025 - 2026, İzmir, Turkey	

- Academic Advisor, *Development of an Artificial Intelligence Infrastructure for Scoring Digital Music Works and Discovering New Music Pieces with High Success Potential*, University-Industry Collaboration Project, 2024 - 2025, İzmir, Turkey
- Principal Investigator, *Solving Asymmetric Social Dilemmas in Multi-Agent Reinforcement Learning with Local Information*, Scientific and Technological Research Council of Turkey (TÜBİTAK), International Postdoctoral Research Fellowship Program for Turkish Citizens, 2024 - 2025, Edinburgh, United Kingdom
- Academic Advisor, *APP2SCALE – AI-Based Auto-Scaling and Fine-Tuning*, University-Industry Collaboration Project, 2023 - 2024, İzmir, Turkey
- Principal Investigator, *Forming Useful Memory in Partially Observable Reinforcement Learning*, Scientific and Technological Research Council of Turkey (TÜBİTAK), Grant No: 120E427, 2021 - 2023, İzmir, Turkey
- Research Fellow, *YÖK Program Ve Kontenjan Yönetim Sistemi (YÖK PKYS)*, Scientific and Technological Research Council of Turkey (TÜBİTAK), Grant No: 115G086, 2018 - 2019, Ankara, Turkey
- Research Fellow, *Subgoal Identification in Sequential Decision Making under Partial Observability*, Scientific and Technological Research Council of Turkey (TÜBİTAK), Grant No: 215E250, 2016 - 2018, Ankara, Turkey

JOURNAL PUBLICATIONS

1. Demir, A., Çilden, E. & Polat, F. Generating Effective Initiation Sets for Subgoal-Driven Options. *Advances in Complex Systems* **22**. doi:10.1142/S0219525919500012 (2019).
2. Demir, A., Çilden, E. & Polat, F. Automatic landmark discovery for learning agents under partial observability. *The Knowledge Engineering Review* **34**. doi:10.1017/S026988891900002X (2019).
3. Demir, A., Çilden, E. & Polat, F. Landmark based guidance for reinforcement learning agents under partial observability. *International Journal of Machine Learning and Cybernetics*, 1–21 (2022).
4. Demir, A. Learning what to memorize: Using intrinsic motivation to form useful memory in partially observable reinforcement learning. *Applied Intelligence*, 1–19 (2023).
5. Uner, B. Y., Demir, A., Zhou, P., Taskiran, E. Z. & Wassenaar, T. Peptide-nanoparticle platforms for antisense therapeutics: A coarse-grained modeling approach to brain delivery. *Computers in Biology and Medicine* **203**, 111479. ISSN: 0010-4825 (2026).

CONFERENCE PROCEEDINGS

6. Demir, A., Çilden, E. & Polat, F. A history tree heuristic to generate better initiation sets for options in reinforcement learning in *Proceedings of the Twenty-second European Conference on Artificial Intelligence (ECAI)* (2016), 1644–1645. doi:10.3233/978-1-61499-672-9-1644.
7. Demir, A., Çilden, E. & Polat, F. Local roots: A tree-based subgoal discovery method to accelerate reinforcement learning in *Joint European Conference on Machine Learning and Knowledge Discovery in Databases (ECML/PKDD)* (2016), 361–376. doi:10.1007/978-3-319-46227-1_23.
8. Demir, A., Çilden, E. & Polat, F. A Concept Filtering Approach for Diverse Density to Discover Subgoals in Reinforcement Learning in *2017 IEEE 29th International Conference on Tools with Artificial Intelligence (ICTAI)* (2017), 1–5. doi:10.1109/ICTAI.2017.00012.
9. Demir, A., Çilden, E. & Polat, F. Landmark Based Reward Shaping in Reinforcement Learning with Hidden States in *Proceedings of the 18th International Conference on Auto-nomous Agents and MultiAgent Systems* (Int. Foundation for Autonomous Agents and Multiagent Systems, 2019), 1922–1924. ISBN: 978-1-4503-6309-9.
10. Dilek, İ., Oğuz, K. & Demir, A. Behaviour Analysis of Izmir Residents Using Public Wi-Fi Access Point Usage in *2022 3rd International Informatics and Software Engineering Conference (IISEC)* (2022), 1–6. doi:10.1109/IISEC56263.2022.9998278.

11. Başoğlu, A. O., Çekirdek, H., Nakipoğlu, K., Yağcı, S. & Demir, A. *Analyzing Traffic Patterns in Izmir: A Study on Busy Hours and Congestion in 2024 Innovations in Intelligent Systems and Applications Conference (ASYU)* (2024), 1–6. doi:10.1109/ASYU62119.2024.10757054.
12. Arslan, A., Titiz, İ. E., Gürcan, E. C. & Demir, A. *Water Consumption Dynamics in Izmir: Analyzing Influences of District, Seasonality, and External Events in 2024 Innovations in Intelligent Systems and Applications Conference (ASYU)* (2024), 1–6. doi:10.1109/ASYU62119.2024.10757161.
13. Çakır, M. T., Nayir, H., Demir, A., Kaya, H. & Ceylan, O. *A reinforcement learning based approach to solve voltage issues in distribution networks in 2025 IEEE 19th International Conference on Compatibility, Power Electronics and Power Engineering (CPE-POWERENG)* (2025), 1–6. doi:10.1109/CPE-POWERENG63314.2025.11027244.
14. Nayir, H., Demir, A. & Kaya, H. *Multi-Dimensional Autoscaling in Microservices Using Reinforcement Learning in 2025 33rd International Symposium on Modeling, Analysis and Sim. of Computer and Telecom. Sys. (MASCOTS)* (2025), 1–8. doi:10.1109/MASCOTS67699.2025.11283269.
15. Demirbilek, B. H., Demir, A. & Polat, F. *External Visual Memory with Autoencoder-Based Intrinsic Motivation for Reinforcement Learning Under Partial Observability in 2025 IEEE 37th International Conference on Tools with Artificial Intelligence (ICTAI)* (2025), 796–800. doi:10.1109/ICTAI66417.2025.00115.
16. Demir, A., Kaya, H., Uluğ, M. & Özkan, I. *Reinforcement Learning for Autoscaling in Cloud Environments: A Case Study from Industry in 2025 16th International Conference on Electrical and Electronics Engineering (ELECO)* (2025), 1–5. doi:10.1109/ELECO69582.2025.11329259.
17. Demir, A., Aydın, H., Tessera, K.-a., Abel, D. & Albrecht, S. V. *Fairness over Equality: Correcting Social Incentives in Asymmetric Sequential Social Dilemmas in Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems* (2026), 13–21. doi:10.65109/CZPZ7833.
18. Uz, E. K., Akgün, H. A., Yontar, H. & Demir, A. *Spatiotemporal Analysis and Incident Severity Forecasting: Unveiling Traffic Patterns in Metropolitan Izmir in 2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)* (2026), 1–9. doi:10.1109/ICHORA69329.2026.11537079.
19. Tur, B., Kobak, E. A., Koğuş, M. & Demir, A. *Temporal and Spatial Dynamics of Public Transit in Istanbul: A Data-Driven Analysis of Passenger Demand Patterns in 2026 8th International Congress on Human-Computer Interaction, Optimization and Robotic Applications (ICHORA)* (2026), 1–8. doi:10.1109/ICHORA69329.2026.11537204.

REFERENCES	Prof. Dr. Faruk Polat, Department of Computer Engineering, Middle East Technical University Ankara, Turkey Dr. Erkin Çilden, STM Defense Technologies Engineering and Trade Inc. Ankara, Turkey Prof. Dr. İsmail Hakkı Toroslu, Department of Computer Engineering, Middle East Technical University Ankara, Turkey Assoc. Prof. Dr. Yusuf Sahillioğlu, Department of Computer Engineering, Middle East Technical University Ankara, Turkey Assoc. Prof. Dr. Sinan Kalkan, Department of Computer Engineering, Middle East Technical University Ankara, Turkey	polat@ceng.metu.edu.tr erkin.cilden@stm.com.tr toroslu@ceng.metu.edu.tr ys@ceng.metu.edu.tr skalkan@ceng.metu.edu.tr
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